

Assignment 2
Deadline – 22nd March 2026

Switched Mode Power Conversion

1. Consider a Boost PFC converter with specifications mentioned below:

Single phase AC voltage input	230 V, 50 Hz
Output DC voltage	400 V
Output voltage ripple	5 % of nominal voltage
Power	1.6 kW
Switching frequency	50 kHz

- a. Design the values boost inductor and output capacitor for the converter. State any assumptions considered during the design. [2]
- b. Determine the blocking voltage and current rating of the semiconductor devices to be used in the converter. [2]
- c. Design the output voltage controller and inductor current controller to regulate the output voltage while maintaining unity power factor from the input AC side. *The output voltage controller should be designed with notch filter at 100 Hz in the voltage feedback.* Report the phase margin and bandwidth of the compensated open loop gains corresponding to the control loops. [6]
- d. Simulate the Boost PFC converter with circuit elements determined in (a) and (b) in MATLAB/Simulink, with closed loop controllers designed in (c). Report the following:
 - i. Steady state AC input voltage, AC input current, output DC voltage and output current at rated load. [4]
 - ii. AC input voltage, AC input current, output DC voltage and output current transient for step load change from 50% to 100% and 100% to 50%. [4]
 - iii. Demonstrate UPF operation, inductor current ripple and output voltage ripple and compare with the design specifications considered in (a). [2]